



Airbrakes: Material Selection & Analysis

Zach Selzman and Metztli Singha

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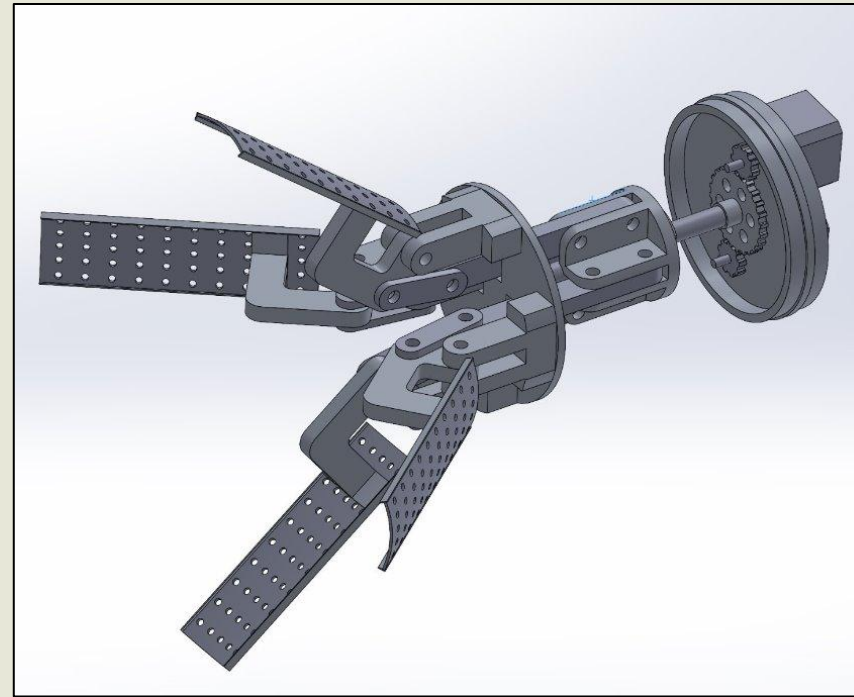
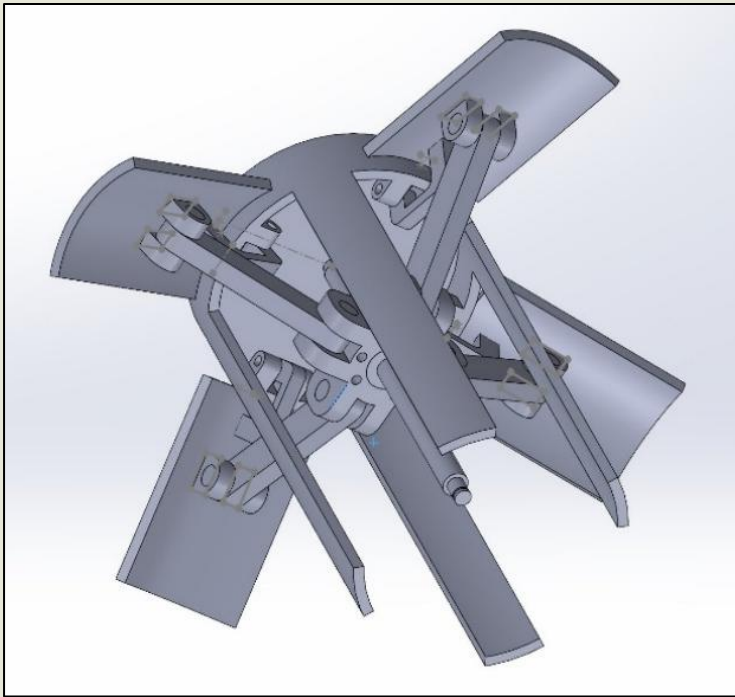
Identification of Design Constraints



Product

Airbrakes

- Slow the velocity of a rocket
- Endure high aerodynamic forces
- Made of 6016-T6 aluminum
- Mill, Lathe, Drill, Water Jet



Stakeholder

Cal Poly Space Systems (CPSS)

- Student-run amateur rocketry club
- Reducing cost and mass are critical
- Reducing environmental impact is secondary
- More “innovative” materials are better for outreach + retention



Needs, Function, Objectives & Constraints

- Need a cheaper/higher performing mechanism to decrease rocket velocity
- Modelled as a panel in bending
 - Stiffness limited
 - Fixed length and width, free thickness
- Aim to minimize cost and mass
- Constraints on yield strength, shear modulus, fracture toughness, heat resistance, and processing



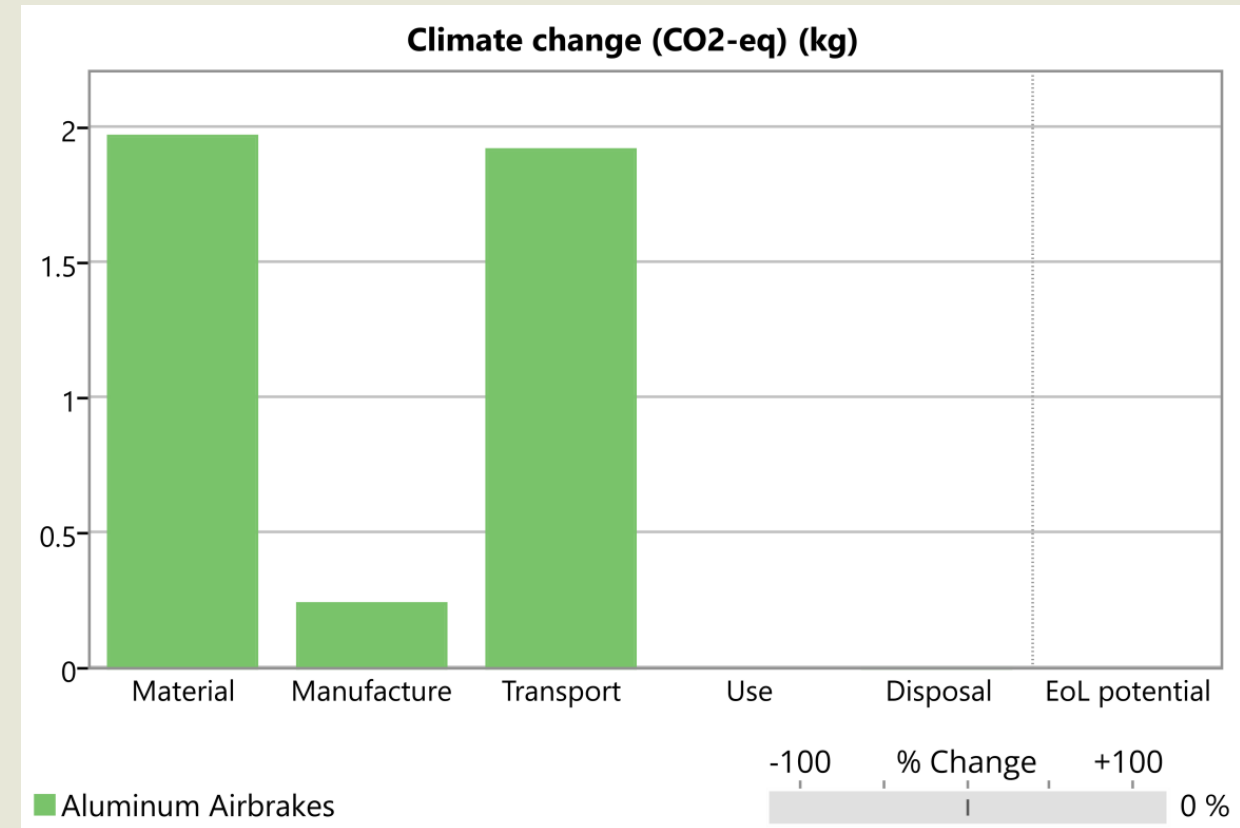
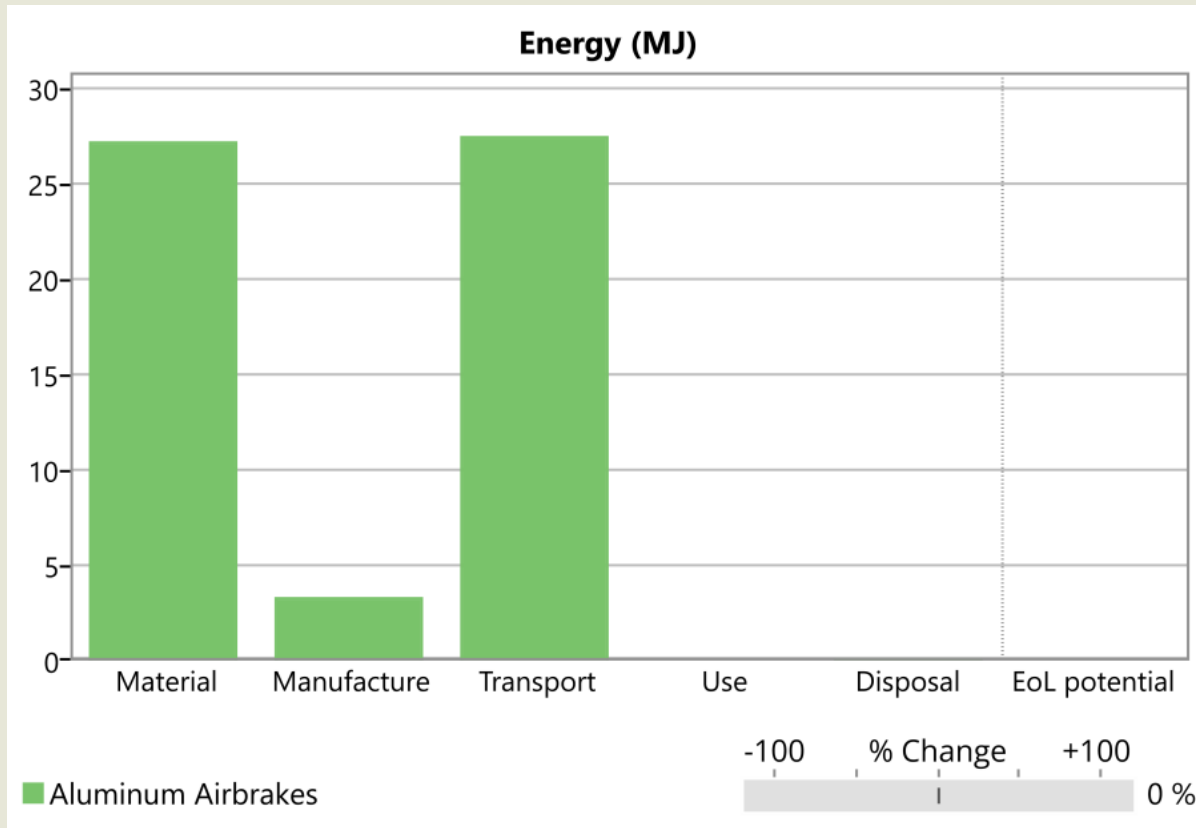
Preliminary Life Cycle Analysis



Preliminary SLCA

Traditionally Manufactured Airbrakes						
	Materials Choice	Energy Use	Solid Residues	Liquid Residues	Gaseous Residues	Total
Premanufacture	2	2	2	2	1	9
Product Manufacture	3	3	2	4	4	16
Product Delivery	4	4	4	4	4	20
Product Use	4	4	4	4	4	20
Recycling, Disposal	4	4	4	4	4	20
Total	17	17	16	18	17	85

Preliminary Eco-Audits



Material Selection



Constraints

Performance

- Yield Strength > 200 MPA
- Fracture Toughness > 20 MPa $\sqrt{\text{m}}$
- Shear Modulus > 15 GPa
- Max Service Temperature > 100 °C

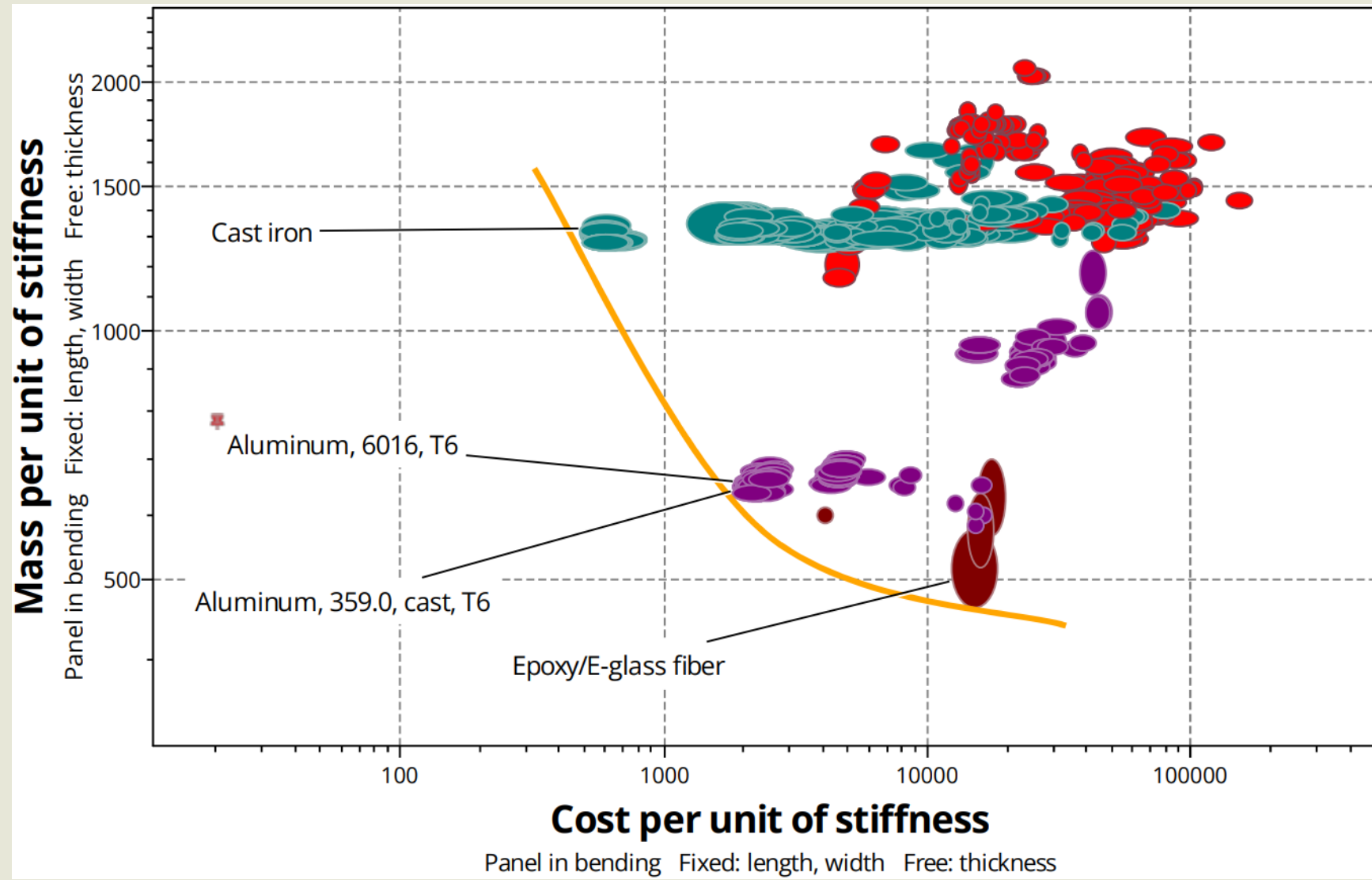
Process

- Manual lathe, mill, water jet, autoclave, saws, drills

Environmental

- Embodied Energy < 348 MJ/kg
- CO2 Emissions < 25.26 kg/kg





Sensitivity Analysis

Two Variations:

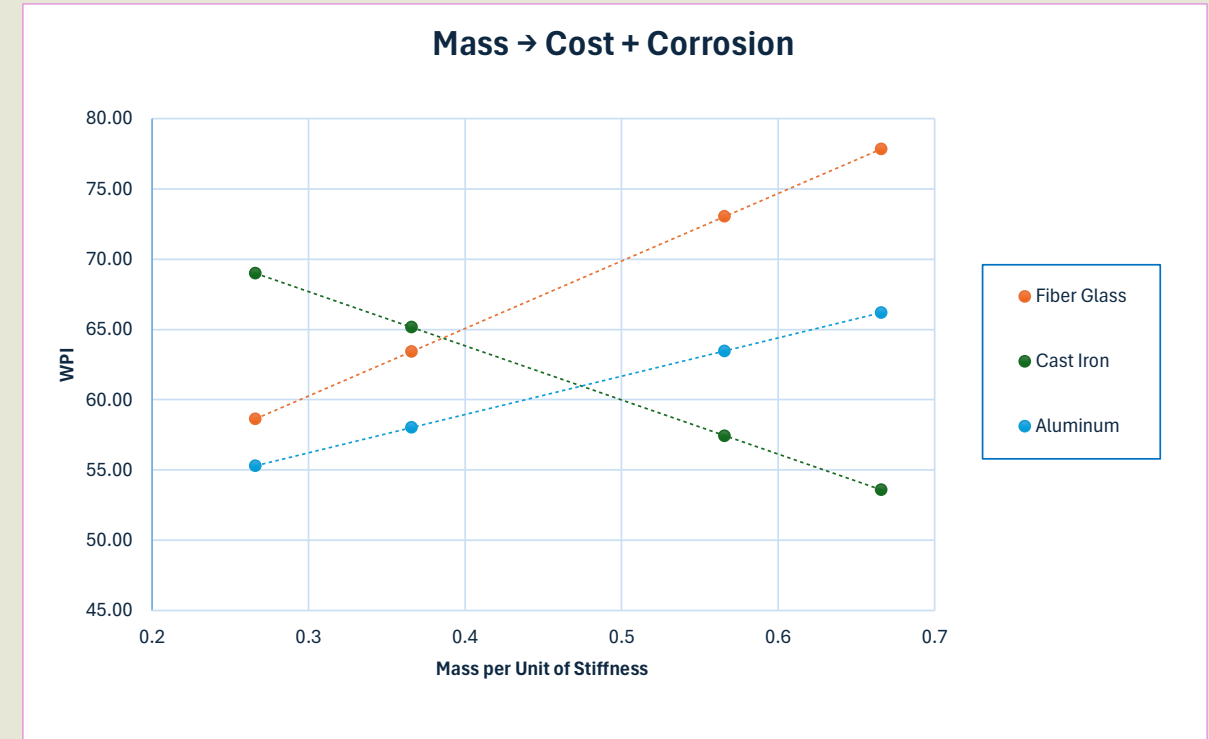
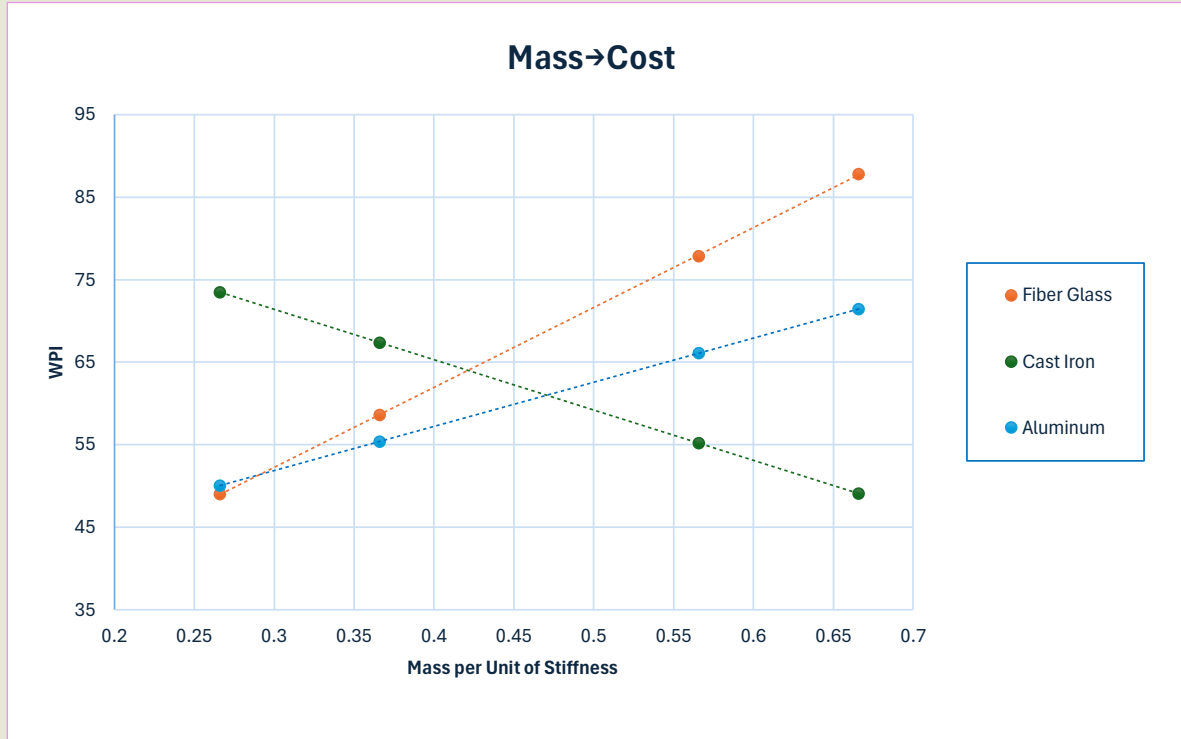
Cost vs. Mass

- Club is poor—cannot exceed budget
- Rocket performance limit is a function of weight
- Direct comparison with the two largest weighting factors

Cost + Corrosion vs. Mass

- Durability is a cost consideration
- Cast Iron mass concerns
- Harsh environment considerations
- Equally distribute the largest weight into the two secondary factors



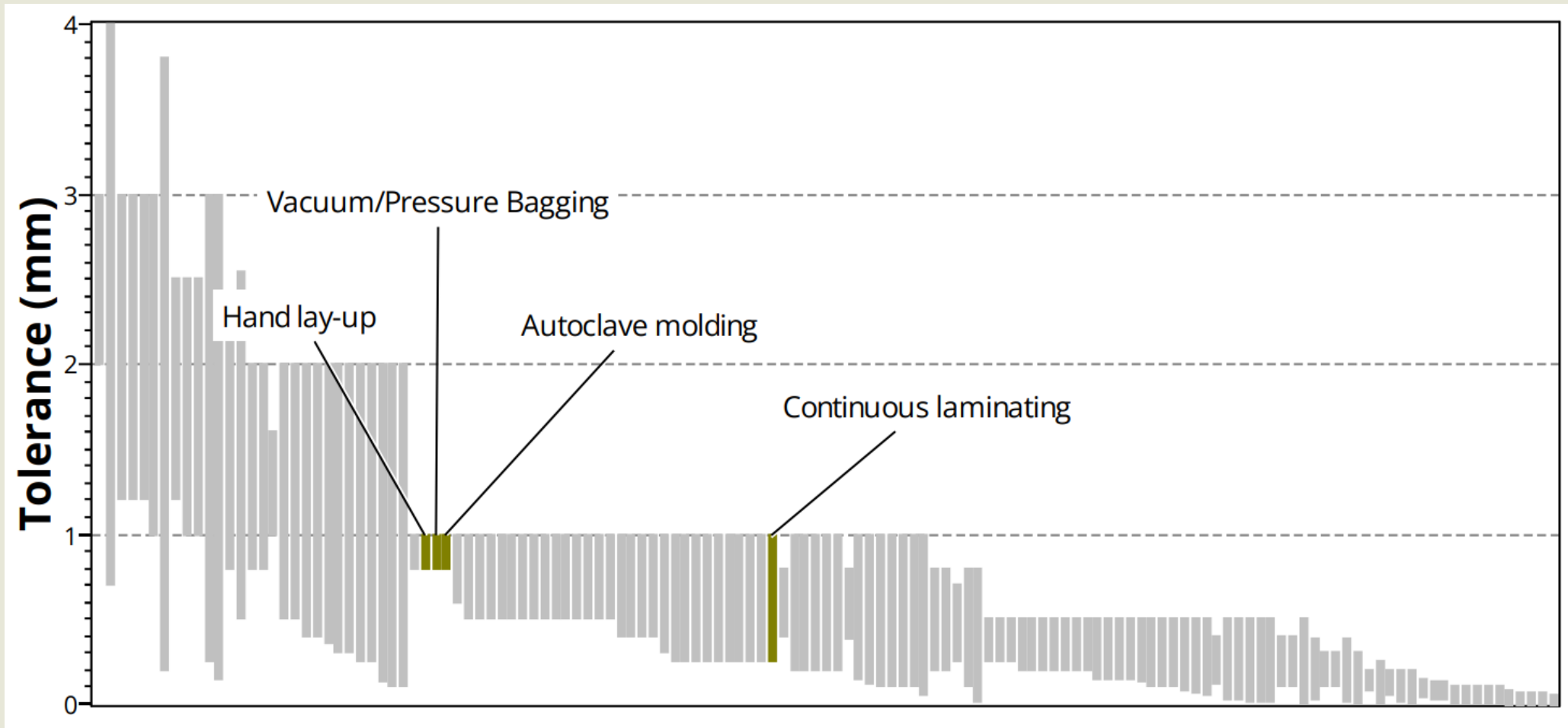


- Aluminum is a “compromise” across the board
- Fiberglass and Cast Iron crossover around 42% with mass into cost
- Delayed until around 39% with corrosion resistance
- Stable solution for use requirements, dubious if the club prioritizes cost
- Tolerance $\sim [0.03, 0.08]$ from weight used (47% mass per unit stiffness)

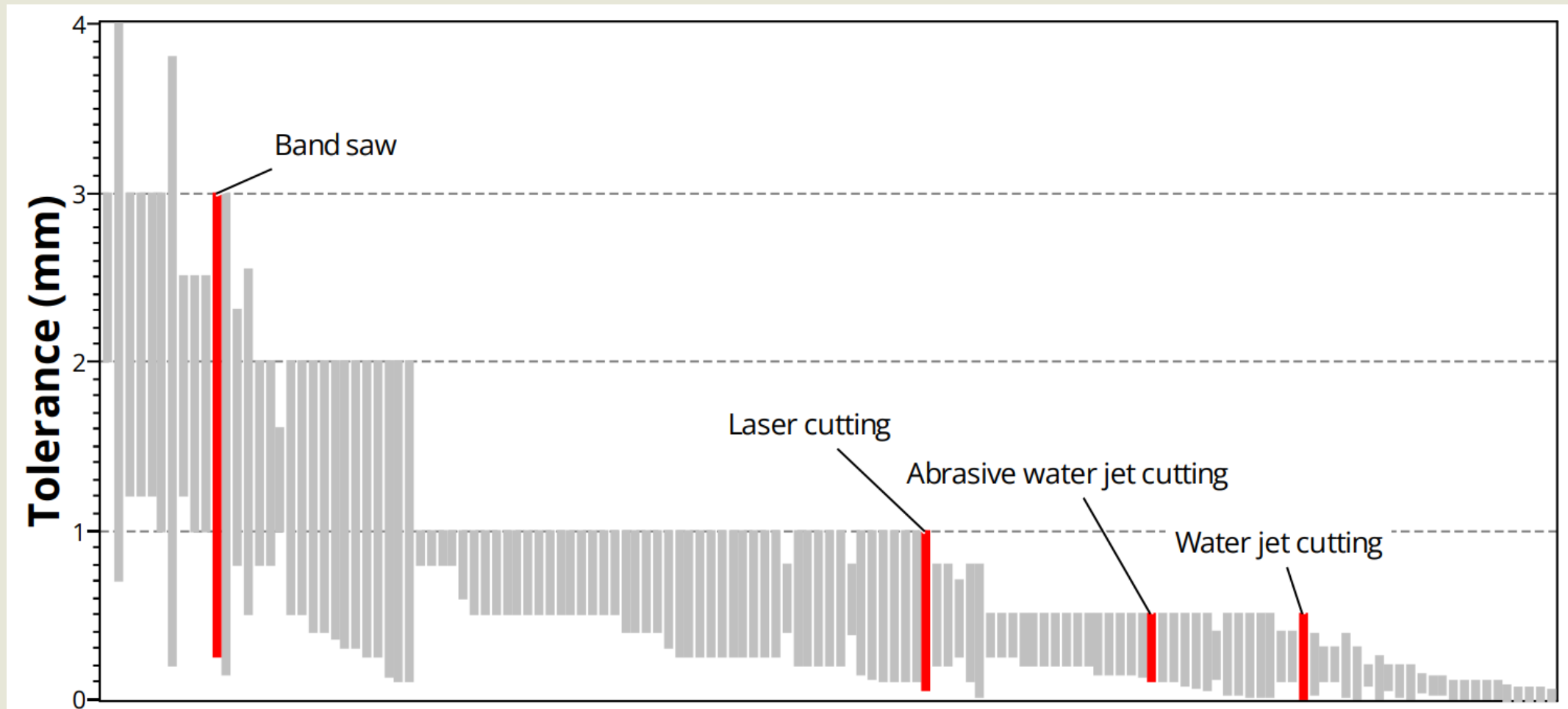
Process Selection



Primary Processes



Secondary Processes



Cost Model



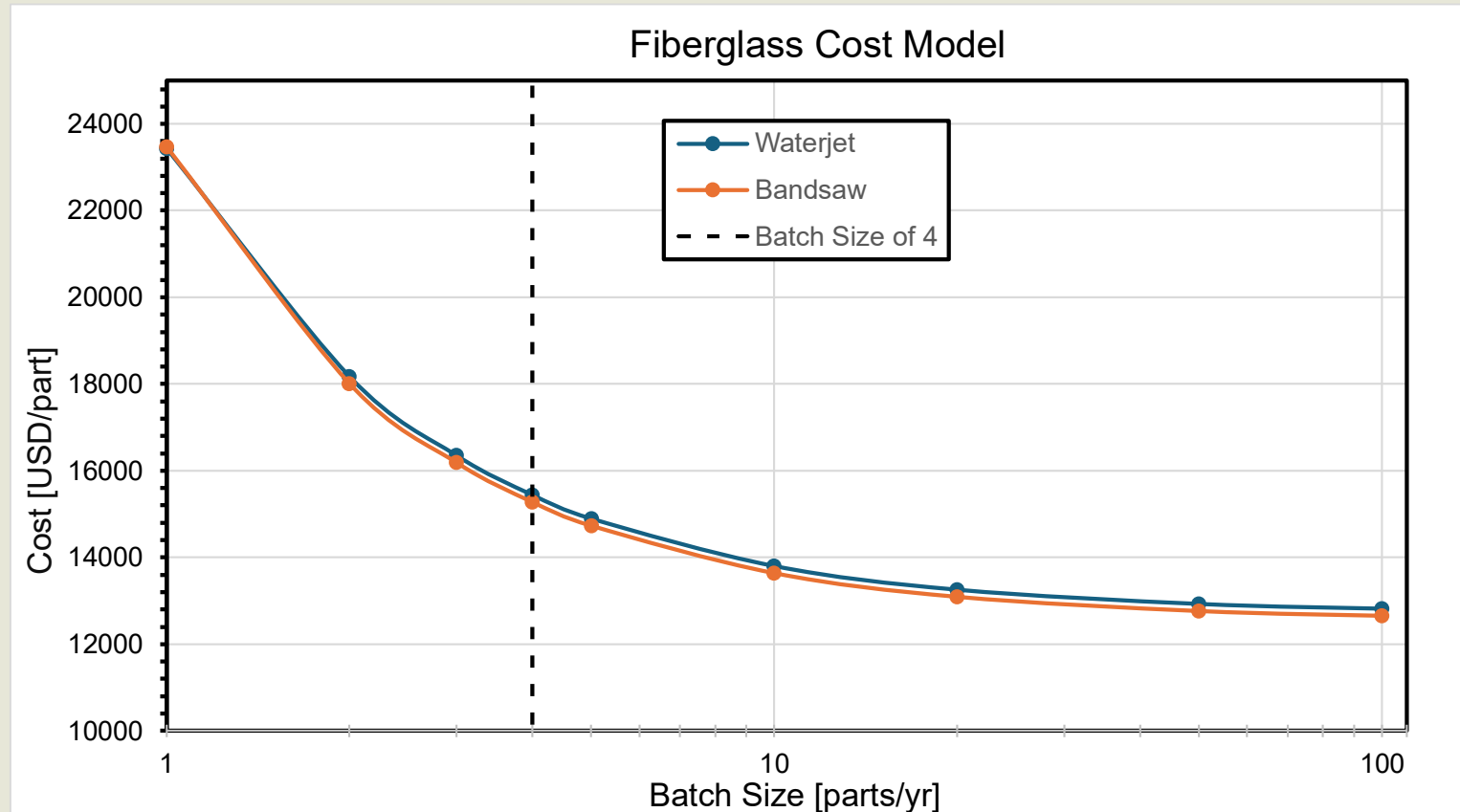
Input Values

	Autoclave Molding	Bandsaw	Waterjet
Material, mCm (\$)	0.964744406	0.964744406	0.964744406
Gross Overhead Rate, CL (\$/h)	600	600	600
Load factor, L	1	1	1
Capital write-off time, two (y)	87600	87600	87600
Discount rate, d	1	1	1
Material Utilization Fraction	0.875	0.875	0.94
Tooling Cost	9921.5	1000	1000
Capital Cost	509200	7395	64700
Batch Rate	0.525	1	6
Years of Use	10	10	10

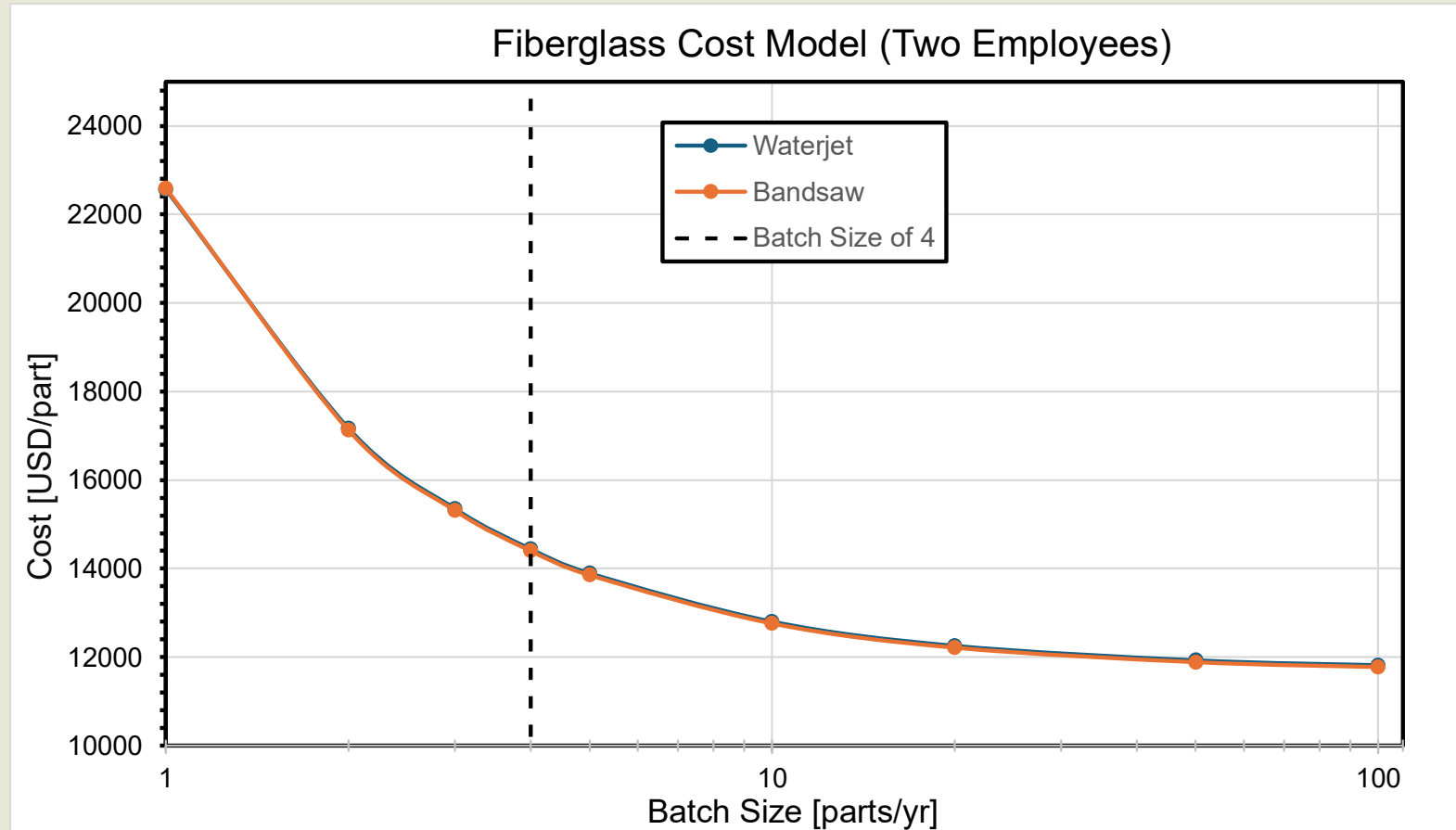
- Assumed to have 10 employees
- Constrained to the machines belonging to the Cal Poly mechanical engineering machine shops

- Values pulled from GrantaEdupack or provided.
- Made assumptions for batch rate and tooling Cost (machine dependent)
- Autoclave is universally used as a primary process, the comparison is between Bandsaw and Waterjet
- Autoclave molding factored as a “fixed cost”
- Waterjet has a much higher capital cost, but a higher material utilization fraction and batch rate





- Expensive overall, largely a result of adding autoclave fixed cost
- At the target batch size, the bandsaw is \$157.41 cheaper than the waterjet
- Can the model be improved?

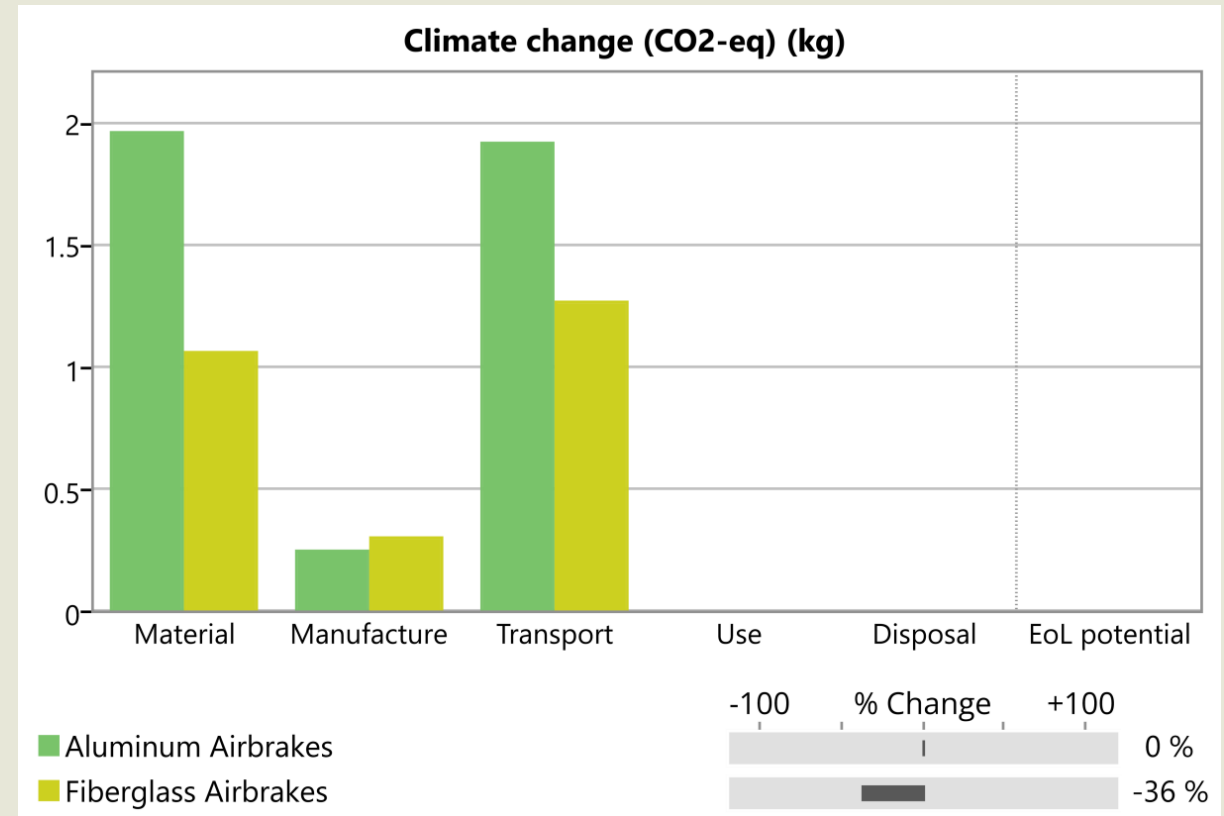
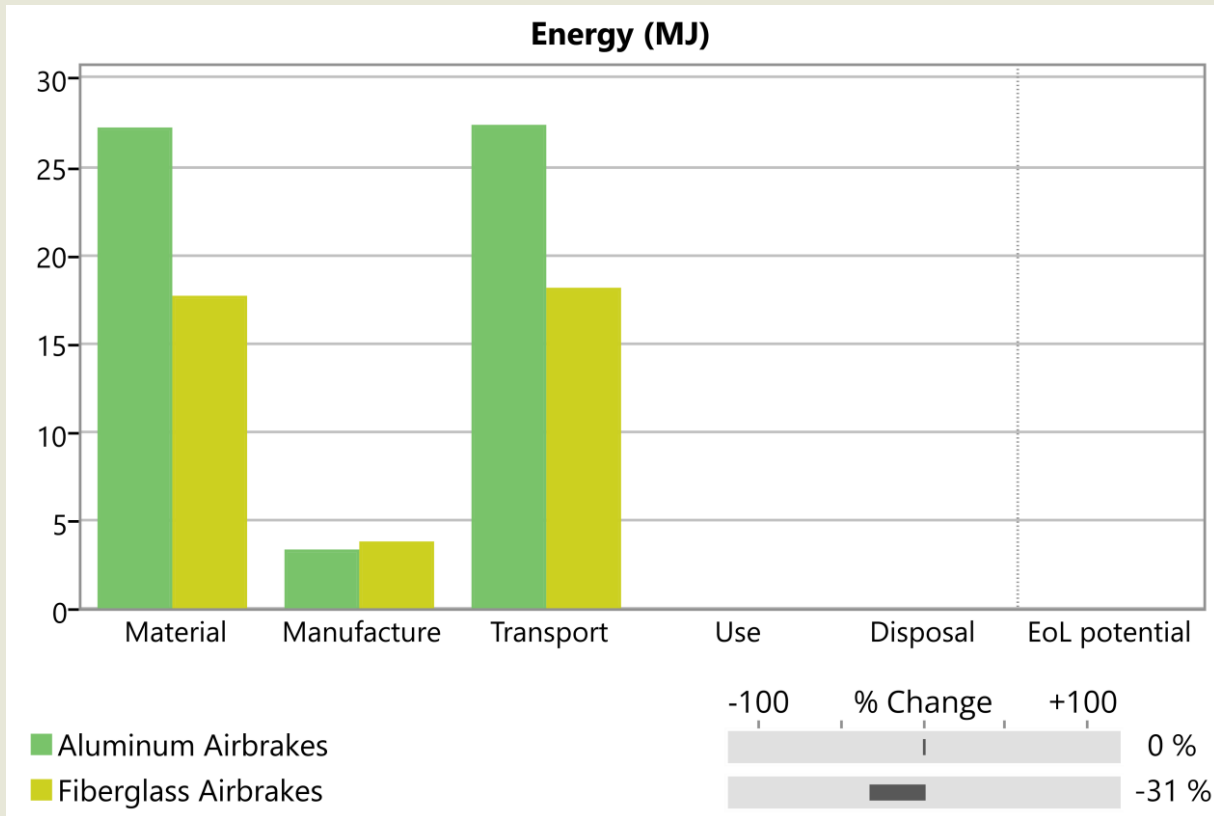


- Overhead rate lowered by reducing the employees from 10 to 2.
- At target batch size, lower Bandsaw cost by \$879.71

Life Cycle Analysis



Fiberglass Eco-Audits

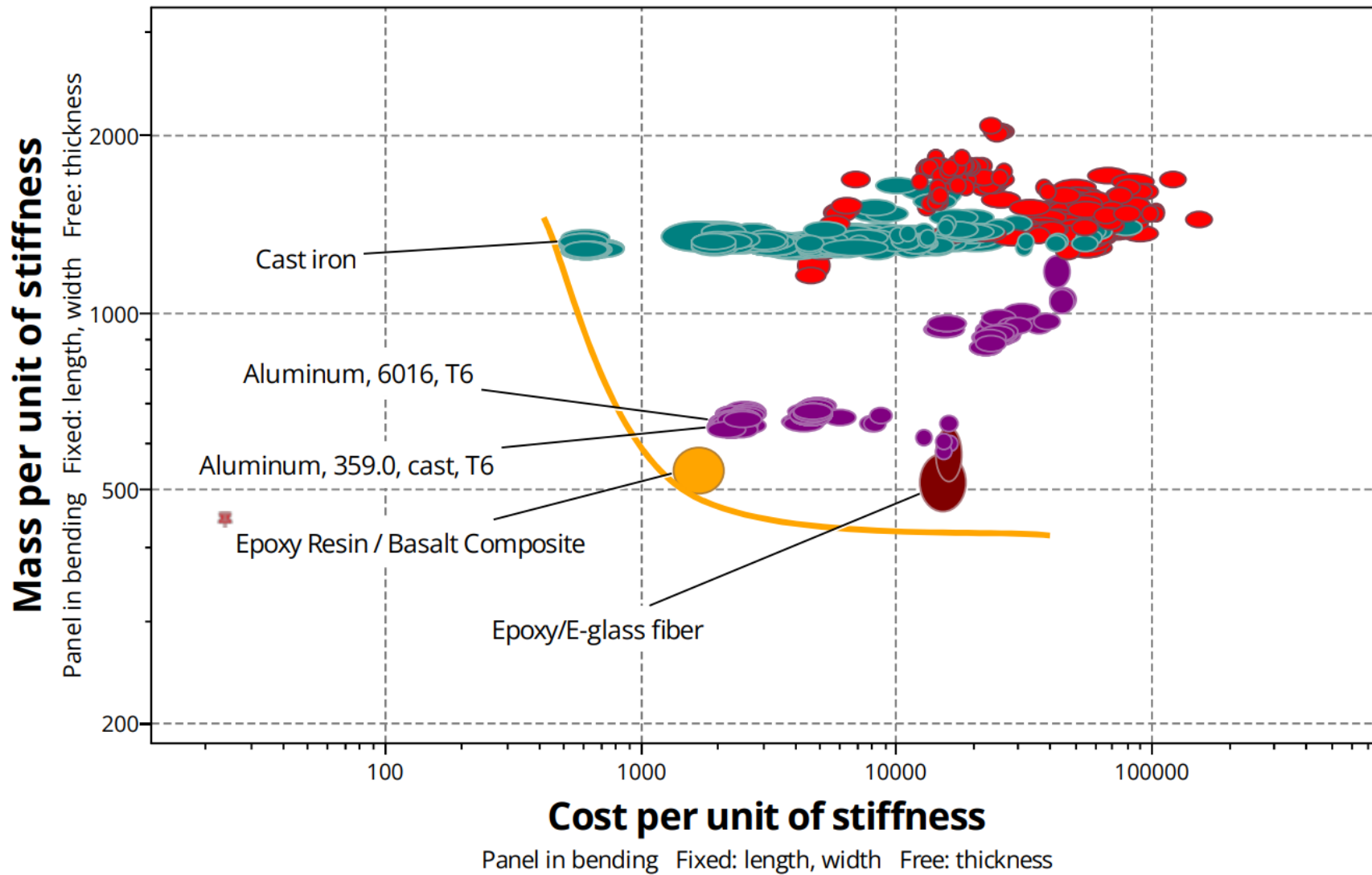


Fiberglass SLCA

Fiberglass Airbrakes						
	Materials Choice	Energy Use	Solid Residues	Liquid Residues	Gaseous Residues	Total
Premanufacture	2	1	2	2	1	8
Product Manufacture	2	1	2	4	2	11
Product Delivery	4	4	4	4	4	20
Product Use	4	4	4	4	4	20
Recycling, Disposal	2	1	0	4	4	11
Total	14	11	12	18	15	70

Improvements





Materials Selection

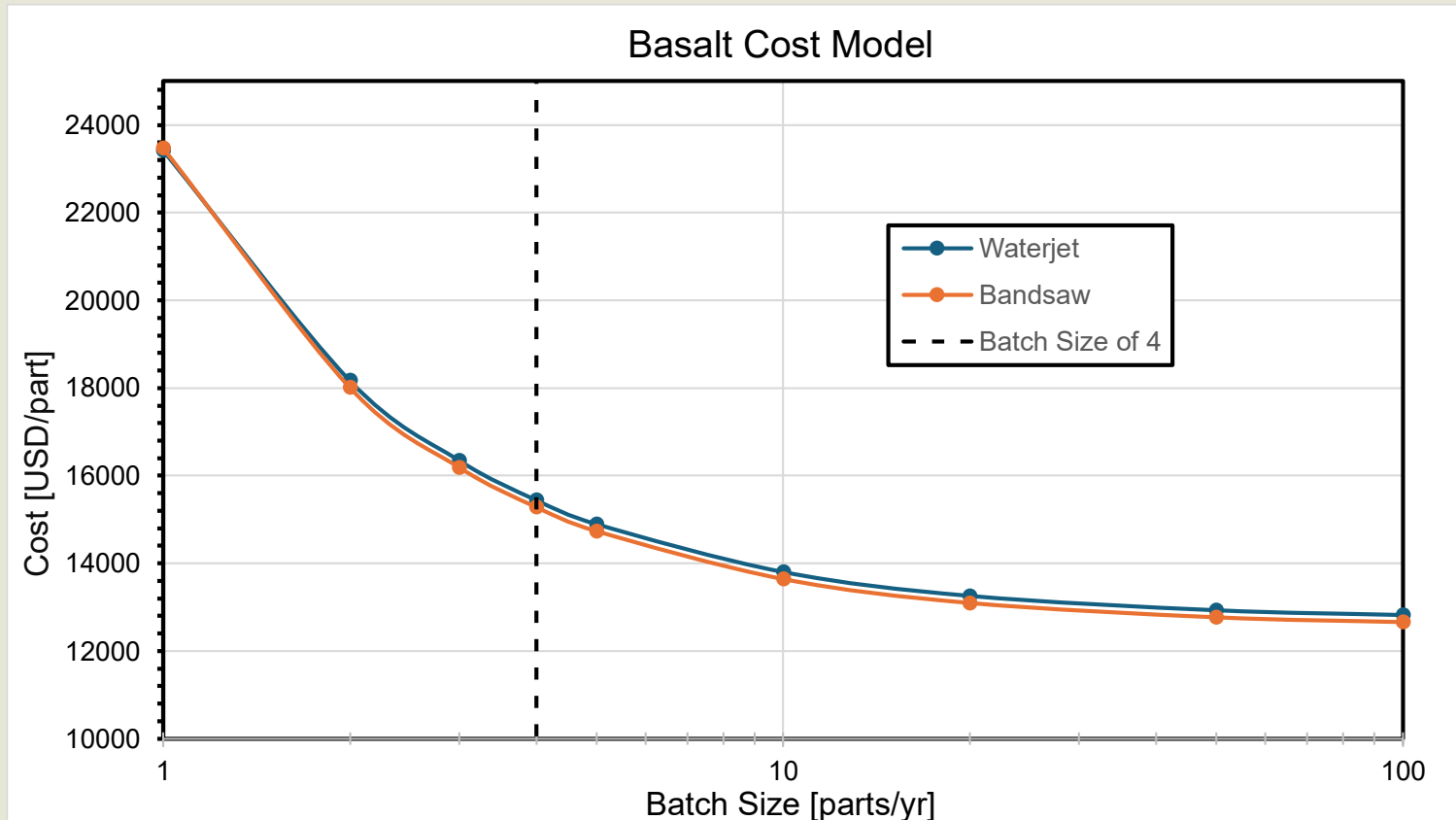
- Basalt had a WPI score of 76.58
- $\Delta WPI = 8.3472 > \text{Fiberglass}$
- Satisfies structural constraints
- Similar Mass/Stiffness to Fiberglass, 10X cheaper



Basalt Cost Model



	Autoclave Molding	Bandsaw	Waterjet
Material, mCm (\$)	0.107916559	0.107916559	0.107916559

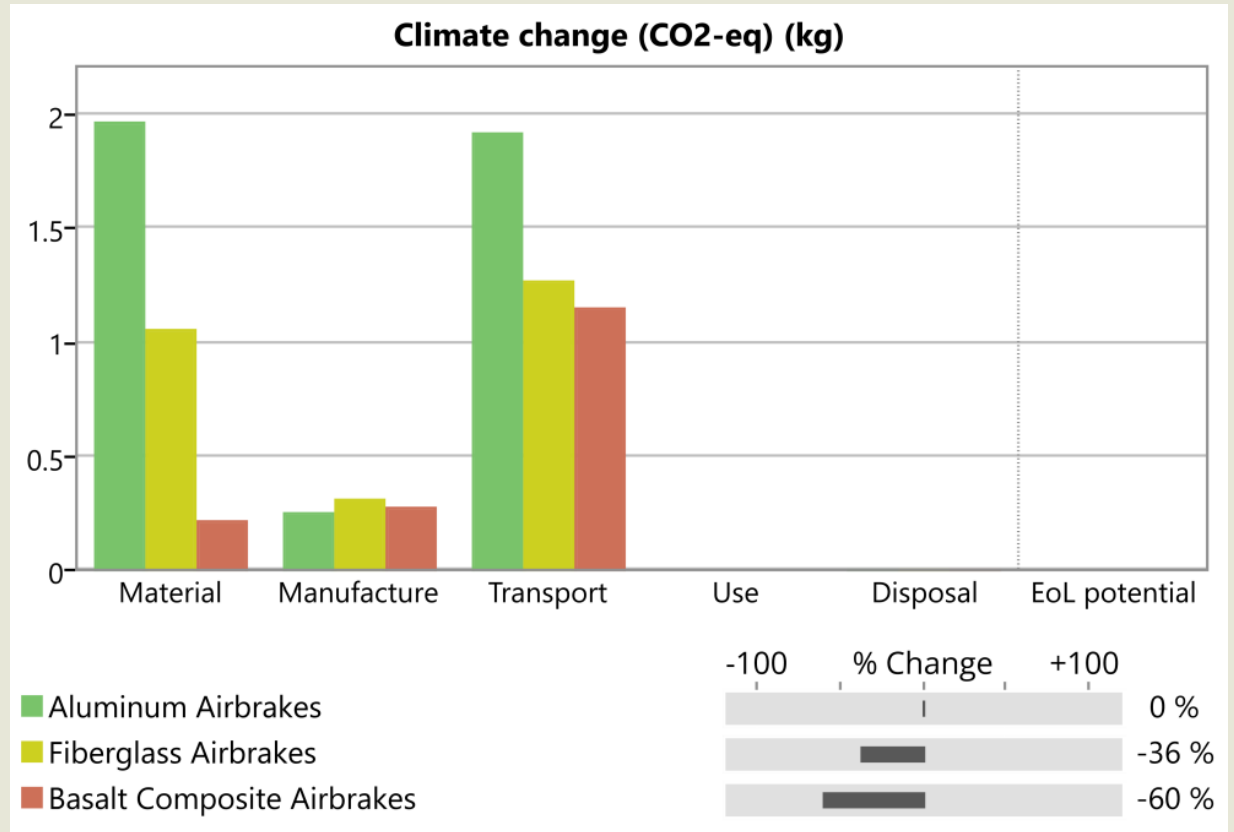
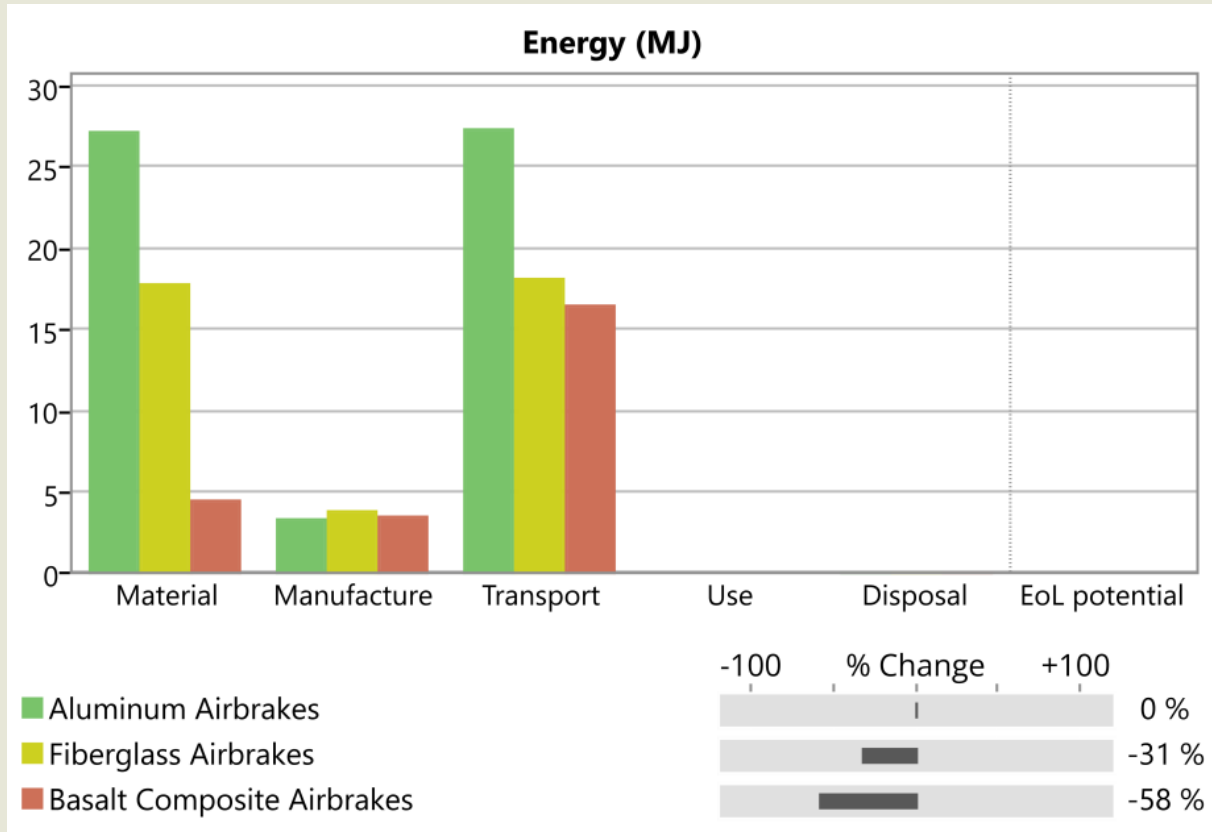


- Shifted graph down, \$157.41 between bandsaw and waterjet (same as Fiberglass)
- At target batch size, \$2 difference between Fiberglass and Basalt
- Material cost is relatively small compared to overhead and tooling

Ecological Analysis



Basalt Eco-Audits



Basalt SLCA

Basalt Airbrakes						
	Materials Choice	Energy Use	Solid Residues	Liquid Residues	Gaseous Residues	Total
Premanufacture	3	2	3	3	2	13
Product Manufacture	3	2	2	3	2	12
Product Delivery	4	4	4	4	4	20
Product Use	4	4	4	4	4	20
Recycling, Disposal	4	4	3	4	4	19
Total	18	16	16	18	16	84

Final Design and Conclusions



- CPSS builds Aluminum air brakes to slow their rockets
- A materials and process selection was completed to select a new material and respective manufacturing method
- Fiberglass, through Autoclave molding and a Bandsaw was selected
- Poor environmental impact led us to creating a 60% filled Basalt-fiber composite in Granta
- Basalt demonstrates significant improvement over Fiberglass, from WPI, cost modeling, and Ecological analysis.
- We recommend 60% filled Basalt Fiber/Epoxy Composite, manufactured with an autoclave and bandsaw.

